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Performance evaluation of subsea blowout preventer systems with common-cause failures

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ABSTRACT

Subsea blowout preventer (BOP) stack configurations are not determined under the present international standards, and both the retrievable pods and non-retrievable pods are used in current projects. This work investigates the effects of stack configurations and mount types on subsea BOP systems from the viewpoint of reliability. A model based on the Markov method for performance evaluation is presented. The subsea BOP system is split into seven independent modules because of its complexity. With respect to common-cause failures, the corresponding Markov models are subsequently developed. The availability and reliability of the system are evaluated by merging the independent Markov models with the Kronecker product approach. The results show that one more subsea ram preventer does not greatly improve the performances of the subsea BOP system, whereas one less subsea annular preventer reduces the performances in the long run. Retrievable control pods markedly improve the performances. Therefore, the subsea BOP system with two annular preventers, four ram preventers, and retrievable pods should be selected preferentially. In addition, the effects of varying of failure rates and mean time to repair for subsea control pods and control stations on the availability and reliability are greater than those for other components, which suggest that the two components should be given more attention when designing subsea BOP systems.

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1. Introduction

A subsea blowout preventer (BOP) system plays an extremely important role in providing safe working conditions for drilling activities in 10,000 ft ultra-deepwater regions. Subsea BOP failures could cause catastrophic accidents such as the explosion of the deep-sea petroleum drilling rig *Deepwater Horizon* and the oil spill off the coast of Louisiana on April 20, 2010. The subsea BOP on the *Deepwater Horizon* rig was believed to have failed isolating the well before and after the explosions. The subsea BOP might have been faulty before the blowout, or it might have been damaged because of the accident (Harlow et al., 2011; Skogdalen et al., 2011). In the wake of recent disasters in oil and gas exploration and production, the performance evaluation of subsea BOP systems is becoming recognized.

The reliability of subsea BOP systems has not been widely studied according to a references review. Shanks et al. studied the reliability of deepwater BOP control system and described a statistical process for determining the reliability and failure rate necessary to accomplish the maintenance goal (Shanks et al., 2003). Holand et al. collected reliability data regarding subsea BOP failures and malfunctions, and calculated the failure rates of subsea BOP components and rig downtime. The availability of the subsea BOP systems was estimated using the fault tree analysis (FTA) method (Holand and Rausand, 1987; Holand, 1991). Fowler et al. studied the system safety of well control equipment including BOPs and hydraulic control systems using failure modes and effects analysis (FMEA) and FTA methods. The results showed that despite human errors, the ram BOP and its associated controls constituted a highly reliable system (Fowler and Roche, 1994).

Aside from FMEA and FTA, other reliability assessment methods such as Petri nets, Markov models, and Bayesian networks have also been used for system reliability analysis. Several authors have indicated that the Markov method is the most suitable for the reliability analysis of redundant systems, such as subsea BOP systems, mainly because of its flexibility (Liu and Rausand, 2011).

Koeppel et al. presented a model for the reliability analysis of supply in multicarrier energy systems based on the Markov method. The models could be used to identify the benefits and sensitivity of certain conversions and their limits in terms of reliability improvements (Koeppel and Andersson, 2009). Cho et al. developed a homogeneous continuous time Markov process model to study the effect of test interval on shutdown system number one in Canadian deuterium–uranium nuclear power plants (Cho and Jiang, 2008). Aksu et al. presented a reliability assessment methodology and its

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application to a combined four-pod propulsion system on a vessel equipped with two fixed-pod and two rotating-pod units using the FMEA, FTA, and Markov methods (Aksu et al., 2006). Veeramany et al. developed a semi-Markov process model to determine the probability of failure of a component cooling water system, which was validated via Monte Carlo simulation (Veeramany and Pandey, 2011). Conn et al. described a generalized semi-Markov model paired with Monte Carlo simulation that represents the evolution of the systems that constitute an offshore installation. The model was used to assess the performance of a maintenance plan in terms of a production penalty and unplanned shutdowns (Conn et al., 2010). Abhulimen presented a model for risk and reliability analysis of complex multifunctional production process system by employing the Monte Carlo and Markov Chain algorithms. The simulation results showed that the risk potential for containment loss from the floating production and storage offloading-riser system for the risk components in parallel exponentially decreased with time as safety ratings increased (Abhulimen, 2009). Based on these studies, the Markov and semi-Markov methods are widely used in the performance evaluations in the nuclear industry, oil and gas industry, and so on.

This paper presents and discusses a Markov-based method developed for the performance evaluation of a subsea BOP system. Based on this developed model, the effects of stack configurations and mounting types of subsea control pods are evaluated. A sensitivity analysis is performed to determine the influences of failure rate and of mean time to repair (MTTR) on the availability and reliability. The remainder of the paper is organized as follows: Section 2 describes the subsea BOP system in detail. Section 3 discusses the Markov model with respect to common-cause failures. Section 4 evaluates the effects of stack configurations, mounting types of subsea control pods, failure rate, and MTTR for each component. Section 5 presents the conclusions.

2. System description

The subsea BOP system consists mainly of the subsea BOP control system and the subsea BOP stack. A typical subsea BOP system is illustrated in Fig. 1.

The subsea BOP control system includes surface and subsea components. The surface components, located in the drilling rig, mainly consist of the central control unit (CCU). It is the kernel of the control system that provides full functional and pressure regulation capability. The CCU is microprocessor-based, and typically utilizes triple modular redundancy (TMR) controllers to transmit commands initiated on the surface to the subsea control pods.

The CCU has three control stations. As the primary control station, driller's panel controls all functions associated with the BOP stack and lower marine riser package (LMRP). The panel design permits well control operations as required, even under adverse conditions. The toolpusher's panel provides the same functionality as the driller's panel, and serves as the secondary control station. It is located in a non-hazardous area away from the drill floor. As the third control station, the work station contains a supervisory control computer and other equipments, such as database servers and network switches.

The subsea components of the control system are two completely independent control pods, the subsea blue pod and the subsea yellow pod, which afford redundant control of all subsea functions. The control pods are the key to system performance. The top section of each pod contains the solenoid-operated shear seal valves and pressure transducers. The lower section contains the subplate-mounted valves, pressure regulators, flow meters, and other associated equipment (Shaughnessy et al., 1999). In addition, each pod includes a subsea electronic module used to receive the command signals initiated on the surface.

Usually, two types of mounting types are used for subsea control pods, retrievable pods and non-retrievable pods (API 16D, 2004). For retrievable pods, surface-controlled locking devices are required to latch the retrievable hydraulic control pods to the LMRP receiver block. Any major problem associated with one pod causes the pod retrieval to the surface for repair, whereas the other one is used to operate all subsea functions, and is unaffected by the disabled one. Non-retrievable pods are the same as the retrievable pods except the pod assembly is fixed to the LMRP. Only when both pods fail are they retrieved for repair.

The configurations of subsea BOP stack vary because of the differences in drilling regions, ocean depths, and so on. Until now, no definite configuration standards have been established.

As secondary barriers during drilling (the primary barrier is the drilling mud), the BOP stack is designed for closing the well annulus or the drill pipe. Two types of preventers, the annular preventer and the ram preventer, are utilized. During deepwater drilling, the BOP stack can be equipped with one or two annular preventers, although usually, the BOP stack is equipped with four or more ram preventers. The different preventer configurations provide different levels of performances for the subsea BOP system. Specifically, a blind shear ram preventer is used to shear the pipe and seal the well, which is regarded as an emergency device. As shown in Fig. 1, the typical subsea BOP stack is equipped with two annular preventers and four ram preventers, including a blind shear ram and three pipe rams.

The subsea BOP stack is usually equipped with two hydraulic connectors, the LMRP connector and the wellhead connector. The LMRP connector is located in the middle of two annular preventers, which is used to connect the LMRP to the BOP stack. The wellhead connector is used to connect the BOP stack to the wellhead. Aside from the annular preventers, ram preventers, connectors, and other components such as the flexible joint, choke/kill valves, and choke/kill lines are integrated to form the whole subsea BOP stack.

To simplify the study on subsea BOP stack reliability, the following assumptions were made:

(1) Only the annular preventers, LMRP connector, ram preventers, and wellhead connector of the subsea BOP stack are investigated; the other components such as the flexible joint and choke/kill lines are neglected because the four main components usually cause subsea BOP stack failures, whereas the other components rarely do.
(2) Failure of the upper annular preventer, subsea control pods, or LMRP connector causes the pulling and repair of the complete subsea BOP stack, although either the LMRP or the complete subsea BOP stack may be pulled for repair depending on different practical situations during actual operation.
(3) The annular preventers or ram preventers are assumed to be pulled for repair only when all preventers fail, although the operators may pull and repair the annular preventers or ram preventers when one, two, three, or even all of them fail, depending on the practical situations.
(4) All failed components of the subsea BOP stack have to be repaired after the stack is pulled to the surface, but not on the seafloor because of their complexity, although some minor failures may be repaired using an underwater robot.

The control stations, control pods, annular preventer, and ram preventer are redundantly configured. Therefore, three control stations, two control pods, several annular preventer, and several ram preventer are considered in parallel. The TMR controllers are considered as a unit because the TMR system is usually supplied as a whole by electronics manufacturing service providers. The whole system can be considered as a series composed of control stations, TMR controllers, subsea control pods, annular preventers, LMRP connector, ram preventer, and wellhead connector because the

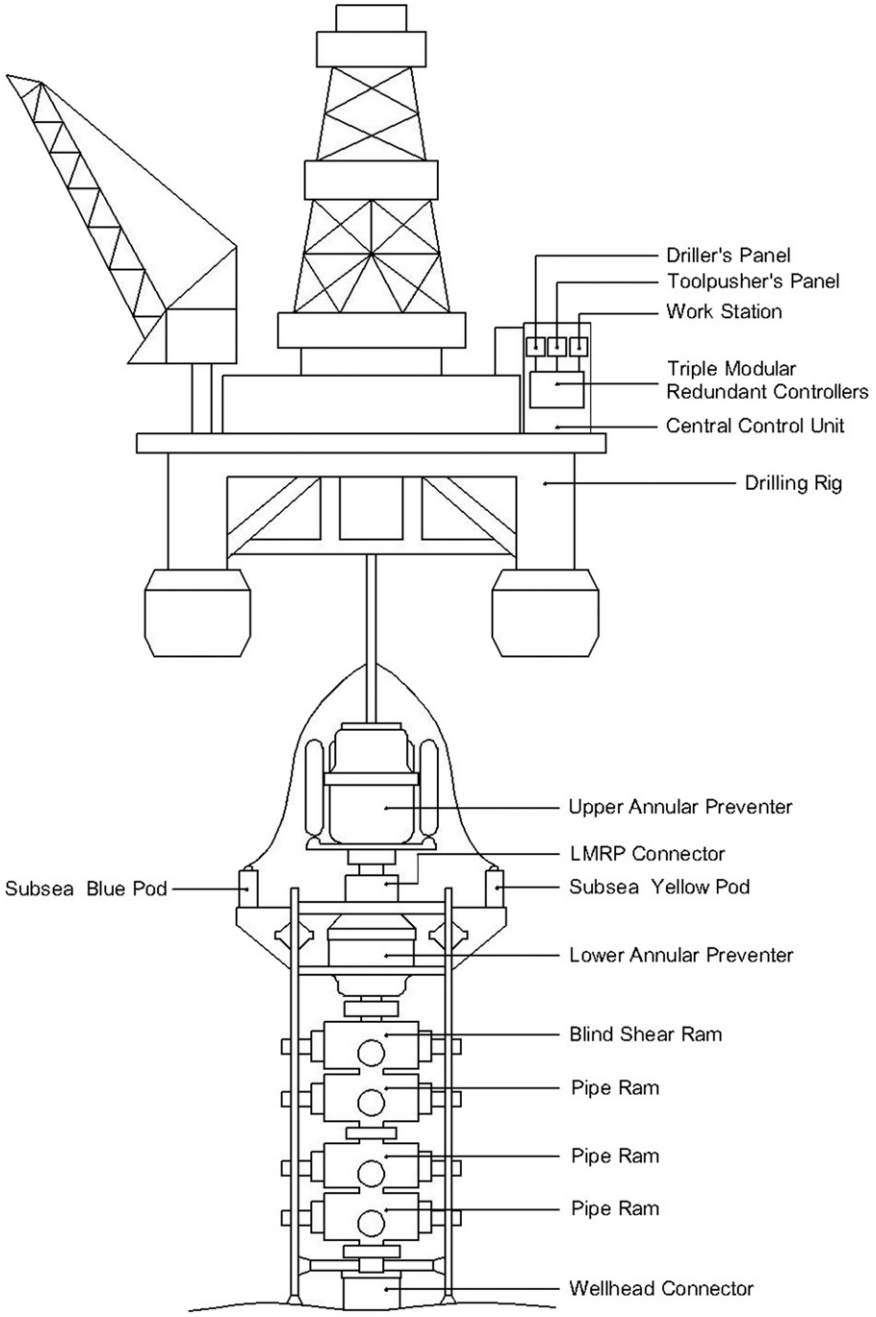


Fig. 1. Typical architecture of a subsea BOP system.

complete failure of each component category causes failure of the subsea BOP system. The relation is represented by a reliability block diagram shown in Fig. 2.

3. Reliability modeling and analysis

3.1. Assumptions

Three subsea BOP system configurations used in deepwater drilling operations were investigated. They are as follows:

(1) BOPA with two annular preventers and four ram preventers;
(2) BOPB with two annular preventers and five ram preventers; and
(3) BOPC with one annular preventer and six ram preventers.

Before developing the models of subsea BOP systems, the following assumptions were made:

(1) The systems start in perfect operation wherein all components are operating correctly;
(2) Both the failure and repair rates are considered constant over time, and are statistically independent;
(3) Components failures can be detected once they fail, and all the failures can be detected, which means that the diagnostic coverage is 100%;
(4) The repaired components are considered as good as new;
(5) For the common-cause failures, only homogeneous components fail simultaneously; heterogeneous components do not fail simultaneously under shock;

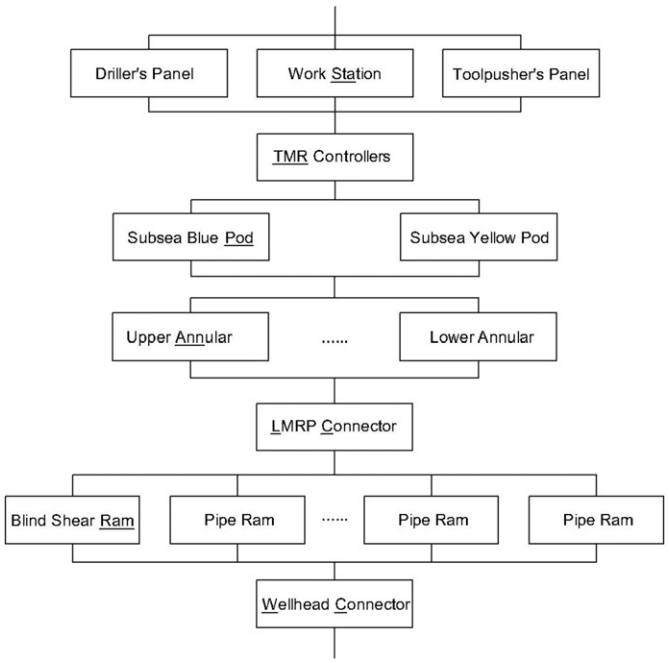


Fig. 2. Reliability block diagram of a subsea BOP system.

(6) For the subsea components, only the failed homogeneous ones are repaired each time, which means that the repair actions for the subsea components are independent.

3.2. Failure rate and repair rate calculations

For redundant system, common-cause failures occur when both redundant components fail from a single event. Experience has shown that common-cause failures have a dominant effect on accidents.

The best known model for common-cause failures is the β-factor model, which has been used widely (IEC 61508, 2000). However, the model is not appropriate distinguishing three or more failures. In this study, the multiple error shock (MESH) model (Hokstad and Bodsberq, 1989) was therefore adopted to describe the common-cause failures of the subsea BOP systems. The model can be used to describe the failure of three or more components under shock.

For the control stations (driller's panel, toolpusher's panel, and work station), the subsea control pods (subsea blue pod and subsea yellow pod), annular preventers (one or two), and ram preventers (four, five, or six), more than one component may fail simultaneously under shock. Based on the MESH model, the statistical mean of the number of components that fail under shock can be expressed as follows:

$$M_I(n) = \sum_{i=1}^{k} i \times p_I(n,i) \tag{1}$$

where I denotes PAN, POD, ANN, and RAM; $p_I(n, k)$ denotes the multiplicity distribution, which is the probability of k components failing with n components in a shock, as shown in Table 1; n is the number of components that work before the shock; k is the number of components that fail in the shock, and $k \le n$. The probability values can be obtained based on various parameter estimation models; however, the estimated values are rather inconsistent or imprecise because they are clearly dependent on the particular system. In this study, the probability values are mainly obtained based on collecting reliability data and engineering judgment.

Table 1
Probability of multiplicity distribution for common-cause failures.

Component	$p(n,2)$	$p(n,3)$	$p(n,4)$	$p(n,5)$	$p(n,6)$
p_{STA}	0.048	0.024	–	–	–
p_{POD}	0.120	–	–	–	–
p_{ANN}	0.083	–	–	–	–
p_{RAM}	0.065	0.012	0.000	0.000	0.000

Table 2
Failure rates of single components.

Component	Failure rate (per h)	Description
λ_{STA}	1.6283×10^{-4}	Failure rates of single control station
λ_{TMR}	8.2547×10^{-6}	Failure rates of TMR controllers
λ_{POD}	1.1433×10^{-4}	Failure rates of single subsea control pod
λ_{ANN}	2.0787×10^{-4}	Failure rates of single annular preventer
λ_{LC}	1.0393×10^{-4}	Failure rates of LMRP connector
λ_{RAM}	1.5590×10^{-4}	Failure rates of single ram preventer
λ_{WC}	1.0393×10^{-4}	Failure rates of wellhead connector

Table 1 indicates that according to the collected reliability data, four or more ram preventers did not fail simultaneously, although it may occur theoretically. Therefore, these probabilities are considered zero.

The relationship between the system shock rate and the individual failure rate for a single component is given by the following equation:

$$\upsilon_I(n) = \frac{n \times \lambda_I}{M_I(n)} \tag{2}$$

The failure rate of a single component is shown in Table 2. These values are obtained based on the collected reliability data and reference review (Holand, 1999, 2001). Only the safety critical failures that occurred after the installation tests and required repair were considered. The failures that occur when the BOP was on the rig, during operation of the BOP, and during the installation tests were not considered in this work.

Therefore, for the control stations, subsea control pods, annular preventers, and ram preventers, the system failure rate for faults of multiplicity k are determined by the system shock rate and the multiplicity distribution probability as follows:

$$\lambda_I(n,k) = \upsilon_I(n) \times p_I(n,k) \tag{3}$$

Given that the repair time was assumed to be exponentially distributed with repair rate μ, the repair rate can be expressed as the reciprocal of MTTR. For the subsea BOP system, the control stations and TMR controllers can be repaired whenever the faults occur because the CCU is located in the drilling rig. Therefore, the repair rate for surface components can be expressed as

$$\mu_X = \frac{1}{k \times MTTR_X} \tag{4}$$

where X denotes STA and TMR, and k is the number of failed components.

However, the retrievable pods or non-retrievable pods have to be pulled and repaired individually or as a whole, respectively. Therefore, the MTTR for the control pods consists of pulling time and repair time. Similarly, the repair actions of annular preventers, ram preventers, and connectors have to be performed by pulling the subsea BOP stack to the surface when the hazardous event occurs. Therefore, the repair rate for subsea components

Table 3
Mean time to repair of single components.

Component	Repair time (h)	Description
$MTTR_{STA}$	7	MTTR of single control station
$MTTR_{TMR}$	10	MTTR of TMR controllers
$MTTR_{POD}$	20	MTTR of single subsea control pod
$MTTR_{ANN}$	48	MTTR of single annular preventer
$MTTR_{LC}$	24	MTTR of LMRP connector
$MTTR_{RAM}$	48	MTTR of single ram preventer
$MTTR_{WC}$	24	MTTR of wellhead connector
$MTTR_{PULL}$	96	MTTR of pulling subsea BOP stack

can be expressed as

$$\mu_Y = \frac{1}{k \times MTTR_Y + MTTR_{PULL}} \tag{5}$$

where Y denotes POD, ANN, LC, RAM, and WC.

Considering its similarity to failure rate, the MTTR for a single component was obtained based on the collected reliability data and reference review (Holand, 1999, 2001), as shown in Table 3.

3.3. Reliability modeling

As shown in Fig. 2, the subsea BOP system is so complex that hundreds of states may exist for the developed Markov model. The creation and solution of such a large Markov model may be very difficult. Based on the assumptions of independent failures (5) and independent repair (6), the BOP system was split into seven manageable and independent modules, namely, the control stations, TMR controller, control pod, annular preventer, LMRP connector, ram preventer, and wellhead connector modules. The corresponding Markov models were subsequently proposed, and the reliability analysis was performed by combining the independent Markov modules with the Kronecker product approach.

Taking the BOPA with retrievable control pods as an example, the section describes subsea BOP system modeling and reliability analysis. Fig. 3 illustrates the state transition diagrams of the seven independent modules. As shown in figure, the failure rates with respect to common-cause failure rate and repair rate are clearly reflected. Specifically, when the subsea BOP control pods were non-retrievable, the repair arc from State 1 to State 0 in Fig. 3(c) was deleted.

Based on the state transition diagrams, the transition rate matrices for the seven modules are given in Eqs. (6)–(12). The elements of the principal diagonal are omitted for the sake of simplicity because the sum of each row is always 1.

$$P_{STA} = \begin{bmatrix} & \lambda_{STA}(3,1) & \lambda_{STA}(3,2) & \lambda_{STA}(3,3) \\ \mu_{STA10} & & \lambda_{STA}(2,1) & \lambda_{STA}(2,2) \\ \mu_{STA20} & 0 & & \lambda_{STA} \\ \mu_{STA30} & 0 & 0 & \end{bmatrix} \tag{6}$$

$$P_{TMR} = \begin{bmatrix} & \lambda_{TMR} \\ \mu_{TMR10} & \end{bmatrix} \tag{7}$$

$$P_{POD} = \begin{bmatrix} & \lambda_{POD}(2,1) & \lambda_{POD}(2,2) \\ \mu_{POD10} & & \lambda_{POD} \\ \mu_{POD20} & 0 & \end{bmatrix} \tag{8}$$

$$P_{ANN} = \begin{bmatrix} & \lambda_{ANN}(2,1) & \lambda_{ANN}(2,2) \\ 0 & & \lambda_{ANN} \\ \mu_{ANN20} & 0 & \end{bmatrix} \tag{9}$$

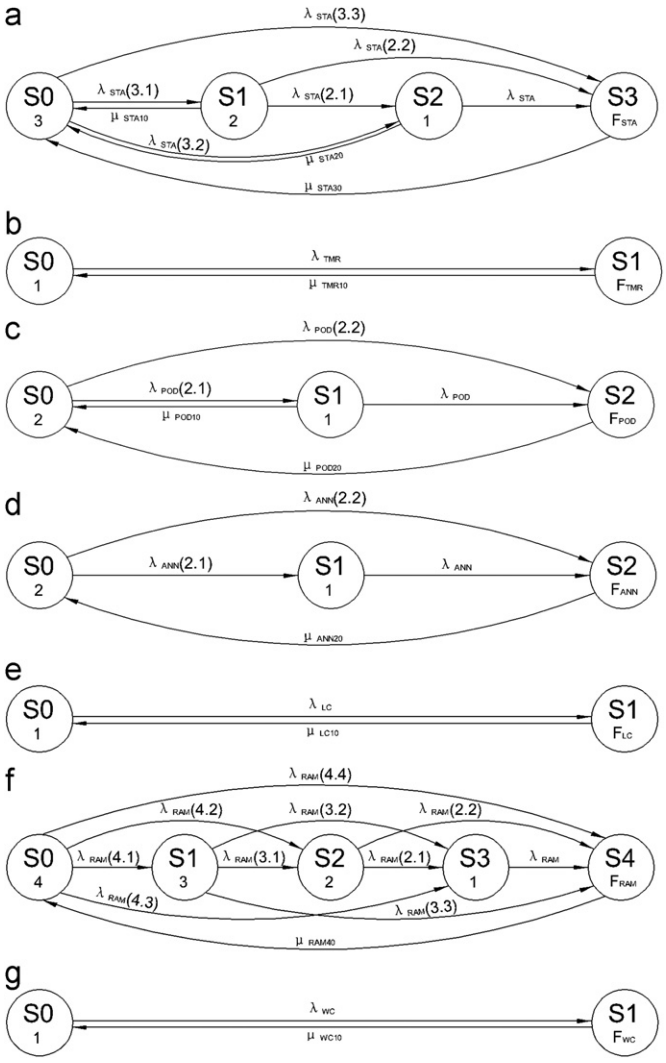


Fig. 3. State transition diagrams of subsea BOPA with retrievable control pods (a) control stations, (b) TMR controllers, (c) control pods, (d) annular preventers, (e) LMRP connector, (f) ram preventers and (g) wellhead connector.

$$P_{LC} = \begin{bmatrix} & \lambda_{LC} \\ \mu_{LC10} & \end{bmatrix} \tag{10}$$

$$P_{RAM} = \begin{bmatrix} & \lambda_{RAM}(4,1) & \lambda_{RAM}(4,2) & \lambda_{RAM}(4,3) & \lambda_{RAM}(4,4) \\ 0 & & \lambda_{RAM}(3,1) & \lambda_{RAM}(3,2) & \lambda_{RAM}(3,3) \\ 0 & 0 & & \lambda_{RAM}(2,1) & \lambda_{RAM}(2,2) \\ 0 & 0 & 0 & & \lambda_{RAM} \\ \mu_{RAM40} & 0 & 0 & 0 & \end{bmatrix} \tag{11}$$

$$P_{WC} = \begin{bmatrix} & \lambda_{WC} \\ \mu_{WC10} & \end{bmatrix} \tag{12}$$

The system state probabilities at specified time t can therefore be written as follows (Dyer, 1989; Goble, 2010):

$$S_J(t) = S_{J0} \times P_J^t \tag{13}$$

where J denotes STA, TMR, POD, ANN, LC, RAM, and WC. S_{J0} is the starting state probability that is a $1 \times n$ matrix. For example, the

starting state probability for the control station module is [1 0 0 0], which suggests that the control stations start in perfect operation.

The transition rate matrix for the whole system was determined using the Kronecker sum of the matrices for the independent modules, whereas the state probability for the whole system was determined using the Kronecker product of the state probabilities for the independent modules (Rausand and Hoyland, 2004). Therefore, the state probability of the subsea BOP system can be expressed as follows:

$$S_{BOP}(t) = S_{STA}(t) \otimes S_{TMR}(t) \otimes S_{POD}(t) \otimes S_{ANN}(t) \otimes S_{LC}(t) \otimes S_{RAM}(t) \otimes S_{WC}(t) \quad (14)$$

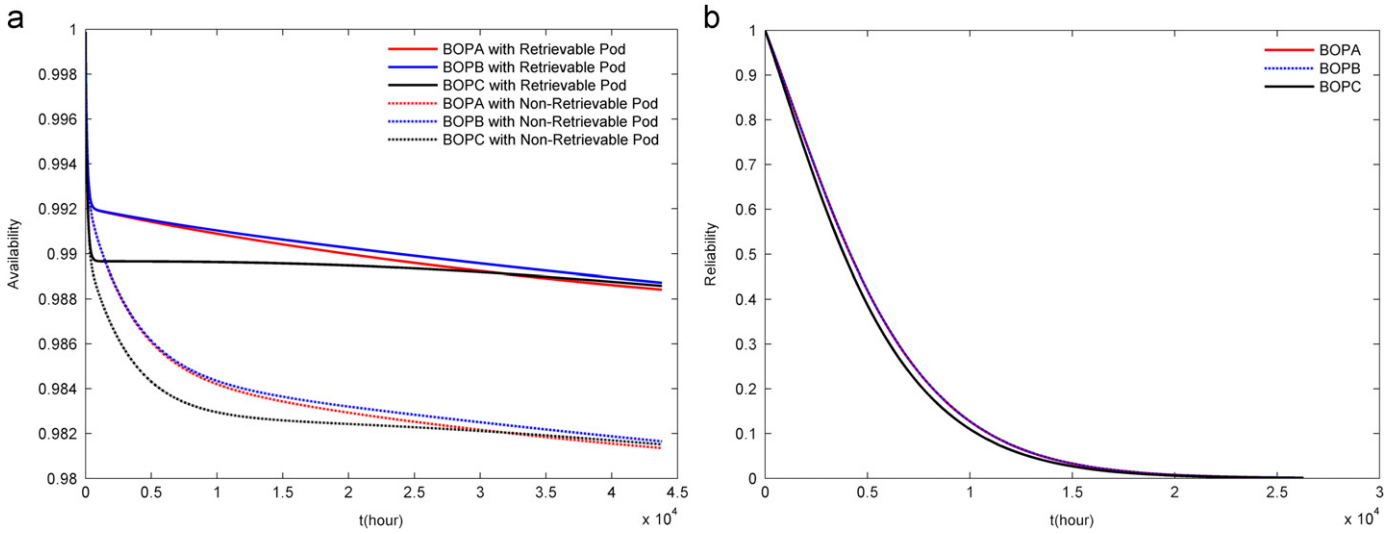


Fig. 4. (a) Availability and (b) reliability for subsea BOP systems with different stack configurations and mounting types of control pods.

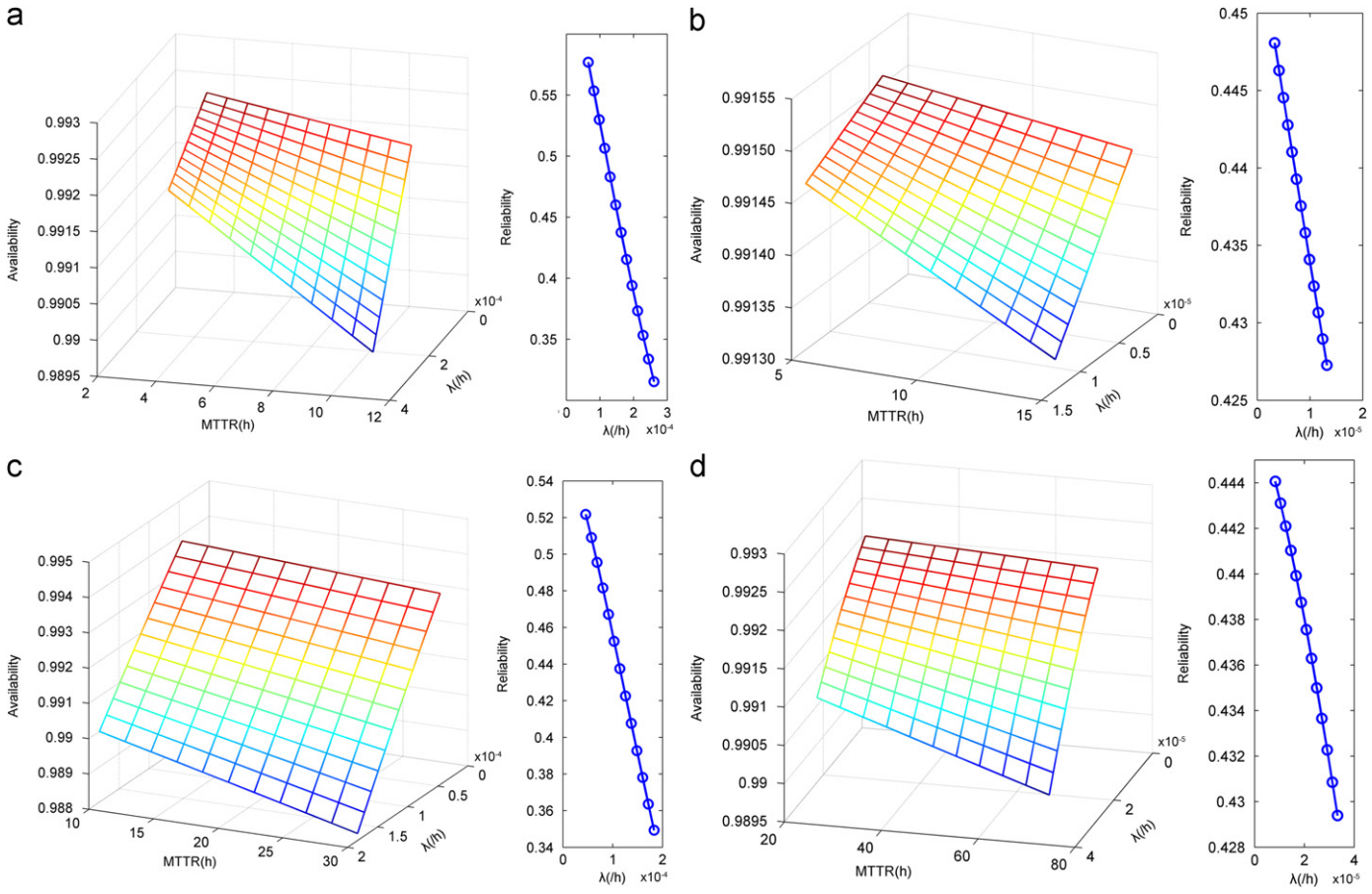


Fig. 5. Effects of failure rates and MTTRs of (a) control stations, (b) TMR controllers, (c) control pods, (d) annular preventers, (e) LMRP connector, (f) ram preventers, (g) wellhead connector on the availability and reliability for subsea BOPA with retrievable pods.

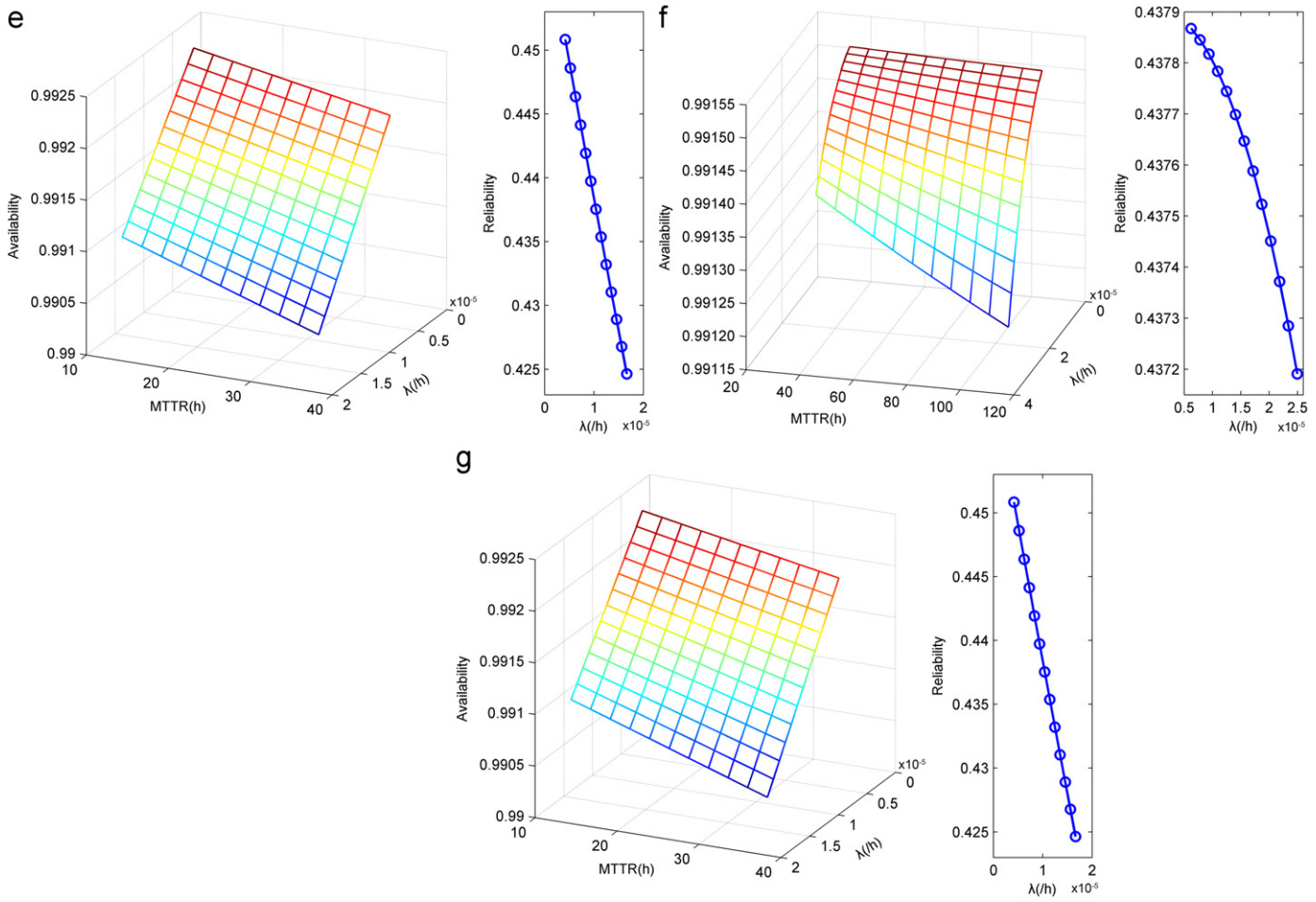


Fig. 5. (*continued*)

3.4. Performance evaluation

For the subsea BOP system, availability and reliability were evaluated.

Availability is the probability that a system or component is performing its required function at a given point in time or over a period of time. In this study, the instantaneous availability of a subsea BOP system at time t was evaluated based on the state probability, and can be expressed as

$$A(t) = S_{BOP}(t) \times V_{BOP} \tag{15}$$

where V_{BOP} is a 1×1 column vector, wherein the corresponding success states are 1, and the corresponding failure states are 0.

Reliability is an important indicator for evaluating non-repairable system; to calculate the reliability of repairable systems, the repair arcs from failure to success states have to be removed. Following the same procedures for availability, the reliability of the subsea BOP system can also be obtained as follows:

$$R(t) = T_{BOP}(t) \times V_{BOP} \tag{16}$$

where T_{BOP} is similar to S_{BOP}, except that it does not contain the repair actions.

4. Results and discussions

Three subsea BOP systems (BOPA, BOPB, and BOPC) with different BOP stack configurations and control pod mounts (retrievable pods and non-retrievable pods) were studied. The availability for a five-year mission time (43,800 h) and the reliability for a three-year mission time (26,280 h) were plotted as shown in Fig. 4.

As indicated in Fig. 4(a), for the subsea BOP systems with retrievable pods, the availability rapidly decreased within the first 400 h, and then, the valves slowly decreased. The availability of BOPB was slightly higher than that of BOPA, and their availabilities were higher than that of BOPC before about 32,000 h. The availability of BOPA becomes lower than that of BOPC after 32,000 h. The results suggest that one more subsea ram preventer did not markedly improve the availability; however, one less subsea annular preventer markedly reduced the availability in the long run. Fortunately, the average drilling time of offshore oil and gas fields is usually less than 200 days (4800 h). The subsea BOP system was overhauled after it was pulled to the surface, which was considered as good as new. Therefore, BOPA is the best choice because it has one less ram preventer than BOPB, but has nearly the same availability. The availability of subsea BOP systems with non-retrievable pods are similar to those with retrievable pods, except the stable values (0.982) for the former are much lower than that for the latter (0.99). The results suggest that the mounts for the control pods greatly affect the performance of subsea BOP systems.

As indicated in Fig. 4(b), the reliability of subsea BOP systems decreases during the course until they reach 0 at about 22,000 h. Obviously, BOPA and BOPB have exactly the same value, which is higher than that of BOPC. Consequently, the repair actions were not considered when system reliability was evaluated. The

mounts have no effect on the reliability of subsea BOP systems. Similar to availability, the reliability analysis also suggests that BOPA is the best choice.

The results of the analysis agree well with the practical situation. For example, the retrievable subsea control pods provide higher availability because of repair actions. BOPB has a higher availability than BOPA because BOPB has one more ram preventer. BOPB provides higher availability and reliability than BOPC because BOPB has a redundant annular preventer. Therefore, the presented Markov model for evaluating the performance of subsea BOP system is considered rational and correct.

To determine the contribution of each component to the availability and reliability of the subsea BOP system, the uncertainty analysis of failure rate and MTTR of each component at 200 h was conducted, as shown in Fig. 5. Each failure rate varies between $\pm 60\%$ from the values given in Table 2, and each MTTR varies between $\pm 50\%$ from the values shown in Table 3. Clearly, for all components with increasing failure rates and MTTR, the availabilities decrease. With increasing failure rates, the reliability also decreases. Therefore, to improve the availability and reliability of the subsea BOP system, both failure rates and MTTR should be reduced.

In addition, Fig. 5 indicated that the influence of each failure rate and MTTR are clearly different. Accounting for the failure rate and MTTR of a single component, the influence on availability is in the following order: subsea control pods, control stations, annular preventers, LMRP connector, wellhead connector, ram preventers, and TMR controllers. The results suggest that retrievable subsea control pods are necessary for the subsea BOP system because they contribute most to availability. The TMR controllers contribute the least because they have a low failure rate. The influence on reliability of the components is in the following order: control stations, subsea control pods, LMRP connector, wellhead connector, TMR controller, annular preventers, and ram preventers. The order is slightly different from that for availability in certain components because of the MTTR of each component. However, the availability is more meaningful because the repair action is obligatory in real work. The analysis results suggest that control stations and subsea control pods contribute considerably to availability and reliability, which should be paid more attention when designing a subsea BOP system.

5. Conclusions

The paper presents a model for the performance evaluation of subsea BOP systems. The model is based on independent Markov submodules merged with the Kronecker product approach. The availability and reliability of three subsea BOP systems with different BOP stack configurations and mounting types of control pods (retrievable pods and non-retrievable pods) are studied. The sensitivity analysis of failure rate and MTTR for each component was performed.

(1) BOPA and BOPB have similar availability and reliability values, whereas these values are higher than those of BOPC, which suggests that one more subsea ram preventer cannot markedly improve the performance of a subsea BOP system, whereas one less subsea annular preventer can reduce the performance for a long period. BOPA with two annular preventers and four ram preventers should be preferred.
(2) The performance of subsea BOP systems with retrievable pods are clearly higher than those with non-retrievable pods, which suggests that the mounts for subsea control pods markedly affect the performance, and the retrievable pods should be preferred.
(3) The influence on failure rate and MTTR of each component on the performance of subsea BOP systems vary widely. The changes in the failure rates and MTTRs of the control pods and control stations markedly affect both availability and reliability. Therefore, the two components should be paid more attention when designing the subsea BOP system.

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References

Abhulimen, K.E., 2009. Model for risk and reliability analysis of complex production systems: application to FPSO flow-riser system. Comput. Chem. Eng. 33, 1306–1321.
Aksu, S., Aksu, S., Turan, O., 2006. Reliability and availability of pod propulsion systems. Qual. Reliab. Eng. Int. 1, 41–58.
API 16D, 2004D. Specification for Control Systems for Drilling Well Control Equipment and Control Systems for Diverter Equipment, second ed. API, Washington, D.C.
Cho, S., Jiang, J., 2008. Analysis of surveillance test interval by Markov process for SDS1 in CANDU nuclear power plants. Reliab. Eng. Syst. Saf. 93, 1–13.
Conn, A.R., Deleris, L.A., Hosking, J.R.M., Thorstensen, T.A., 2010. A simulation model for improving the maintenance of high cost systems, with application to an offshore oil installation. Qual. Reliab. Eng. Int. 26, 733–748.
Dyer, D., 1989. Unification of reliability/availability/repairability models for Markov systems. IEEE Trans. Reliab. 2, 246–252.
Fowler, J.H., Roche, J.R., 1994. System safety analysis of well-control equipment. SPE Drilling Completion 3, 193–198.
Goble, W.M., 2010. Control Systems Safety Evaluation and Reliability, third ed. ISA, North Carolina.
Harlow, W.F., Brantley, B.C., Harlow, R.M., 2011. BP initial image repair strategies after the Deepwater Horizon spill. Public Relat. Rev. 37, 80–83.
Hokstad, P., Bodsberq, L., 1989. Reliability model for computerized safety systems. Proc. Annu. Reliab. Maintainability Symp., 435–440.
Holand, P., 1991. Subsea blowout-preventer systems reliability and testing. SPE Drilling Eng. 4, 293–298.
Holand, P., 1999. Reliability of Subsea BOP Systems for Deepwater Application, Phase II DW. Project Report STF38 A99426.
Holand, P., 2001. Deepwater Kicks and BOP Performance. Project Report STF38 A01419.
Holand, P., Rausand, M., 1987. Reliability of subsea BOP systems. Reliab. Eng. 19, 263–275.
IEC 61508, 2000. Functional Safety of Electrical/Electronic/Programmable Electronic: Status Related System. IEC, Geneva.
Koeppel, G., Andersson, G., 2009. Reliabiltiy modelling of multi-carrier energy systems. Energy 34, 235–244.
Liu, Y.L., Rausand, M., 2011. Reliability assessment of safety instrumented systems subject to different demand modes. J. Loss Prev. Process Ind. 24, 49–56.
Rausand, M., Hoyland, A., 2004. System Reliability Theory, Models, Statistical Methods, and Applications, second ed. Wiley, Hoboken, New Jersey.
Shanks, E., Dykes, A., Quilici, M., Pruitt, J., 2003. Deepwater BOP Control System–A Look at Reliability Issues. Offshore Technology Conference, 1–10 pp.
Shaughnessy, J.M., Armagost, W.K., Crop, A., Herrmann, R.P., Cleaver, C.M.A., 1999. Problems of ultra-deepwater drilling. In: Proceedings of the IADC SPE Asia Pacific Drilling Technology APDT, vol. 1, 179–188 pp.
Skogdalen, J.E., Utne, I.B., Vinnem, J.E., 2011. Developing safety indicators for preventing offshore oil and gas deepwater drilling blowouts. Saf. Sci. 48, 1187–1199.
Veeramany, A., Pandey, M.D., 2011. Reliability analysis of nuclear component cooling water system using semi-Markov process model. Nucl. Eng. Des. 24, 1799–1806.